

Class 9 Physics: Motion

Comprehensive Board Exam Preparation & Formula Guide

Subject: Physics (CBSE Class 9)

Chapter: Motion

Target: Board Exams & Foundation

1. Core Concepts & Definitions

Rest and Motion

Motion is a change in position of an object with respect to time and a reference point (origin). Rest and motion are relative terms; an object can be at rest relative to one frame of reference while in motion relative to another.

Distance vs. Displacement

Property	Distance	Displacement
Definition	Actual length of the path traveled by a body.	Shortest straight-line distance from initial to final position.
Nature	Scalar quantity (Magnitude only).	Vector quantity (Magnitude and Direction).
Value	Always positive (> 0 for moving body).	Can be positive, negative, or zero.
SI Unit	Meter (m)	Meter (m)

2. Speed, Velocity & Acceleration

Speed: Distance covered per unit time. Scalar quantity. SI unit: m/s .

$$\text{Speed } (v) = s / t \quad | \quad \text{Average Speed} = \text{Total Distance} / \text{Total Time}$$

Velocity: Displacement per unit time (Speed in a given direction). Vector quantity. SI unit: m/s .

$$\text{Average Velocity } (v_{avg}) = (u + v) / 2 \quad (\text{for uniform acceleration})$$

Acceleration: Rate of change of velocity with respect to time. Vector quantity. SI unit: m/s^2 .

$$a = (v - u) / t$$

- **Positive Acceleration:** Velocity increases with time ($v > u$).
- **Negative Acceleration (Retardation):** Velocity decreases with time ($v < u$).

3. Key Equations of Motion (Uniform Acceleration)

1. **First Equation (Velocity-Time):** $v = u + at$
2. **Second Equation (Position-Time):** $s = ut + \frac{1}{2}at^2$
3. **Third Equation (Position-Velocity):** $v^2 - u^2 = 2as$

Where: u = initial velocity, v = final velocity, a = uniform acceleration, t = time elapsed, s = displacement.

4. Graphical Interpretation

- **Distance-Time Graph ($s-t$):** The slope of the distance-time graph gives the **Speed** of the object.
- **Velocity-Time Graph ($v-t$):**
 - The slope of the velocity-time graph gives the **Acceleration**.
 - The area under the curve of the velocity-time graph gives the total **Displacement / Distance**.

5. Uniform Circular Motion

When an object moves along a circular path with a constant speed, its direction continuously changes, resulting in a continuous change in velocity. Hence, uniform circular motion is an **accelerated motion**.

$$\text{Speed } (v) = (2 \pi r) / T$$

(Where r = radius of the circle, T = time for one complete revolution)

6. Board Exam Solved Practice Questions

Numerical • 3 Marks

Q1. A bus decreases its speed from 80 km/h to 60 km/h in 5 seconds. Find the acceleration of the bus.

Solution:

Initial velocity (u) = 80 km/h = $80 \times (5 / 18) = 22.22$ m/s

Final velocity (v) = 60 km/h = $60 \times (5 / 18) = 16.67$ m/s

Time (t) = 5 s

Acceleration (a) = $(v - u) / t = (16.67 - 22.22) / 5 = -5.55 / 5 = -1.11 \text{ m/s}^2$

(The negative sign indicates retardation/deceleration).

Numerical • 3 Marks

Q2. A train starting from rest attains a velocity of 72 km/h in 5 minutes. Assuming uniform acceleration, find: (i) the acceleration, (ii) the distance traveled in this time.

Solution:

$u = 0$ m/s, $v = 72$ km/h = 20 m/s, $t = 5$ min = 300 s.

(i) $a = (v - u) / t = (20 - 0) / 300 = 1/15 \text{ m/s}^2$ (or 0.067 m/s^2)

(ii) Using $v^2 - u^2 = 2as$:

$(20)^2 - 0 = 2 \times (1/15) \times s \Rightarrow 400 = (2/15)s \Rightarrow s = (400 \times 15) / 2 = 3000 \text{ m (3 km)}.$

Conceptual • 2 Marks

Q3. Under what condition is the magnitude of average velocity equal to average speed?

Answer: When an object moves along a straight line in a single direction without turning back, the distance traveled is equal to the displacement. In this case, average speed equals average velocity.